INCH-POUND

A-A-59127
October 15, 1997

COMMERCIAL ITEM DESCRIPTION

Generator, Signal, VOR/ILS, AN/ARM-180

The General Services Administration has authorized
the use of this commercial item description (CID) for all federal agencies.

l. SCOPE. This commercial item description covers a VOR/ILS signal generator. This CID is meant as a minimum requirement for the VOR/ILS signal generator in which only those manufacturers that meet or surpass the following requirements are qualified per this CID. This signal generator shall produce both rf and audio signals for testing VOR, localizer, glideslope, and marker beacon signals.

2. SALIENT CHARACTERISTICS.

2.1 The equipment shall meet Class 3 requirements of MIL-PRF-28800 and be bid sample tested within the accuracies, limits, and specifications herein.

2.2 Environmental requirements.

2.2.1 Temperature requirements. The product shall meet its performance and accuracy requirements in an operating temperature of +10°C to +40°C and non-operating -20°C to +65°C.

2.2.2 Electromagnetic interference (EMI). The equipment shall comply with the EMI requirements specified for Class B equipment in MIL-STD-461, excluding tests CE06, CE07 and modify tests RE02, RS03 as follows.

a. RE02 full limits to 10 times the highest used or intentionally generated frequency or 1 GHz, whichever is greater.

b. RS03 full limits to 1 GHz.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Technology & Industrial Services Division, SA-ALC/TILDD, Bldg. 171, 485 Quentin Roosevelt Rd., Kelly AFB, TX 78241-6524.

AMSC N/A FSC 6625

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2.3 Safety requirements. The equipment shall meet safety requirements of MIL-PRF-28800.

2.4 Electrical power source. The product shall be switchable to operate from commercial, military, and shipboard power sources of 115/230 Volts AC (± 10%), from 50 to 60 Hz. The power input shall be recess mounted through the rear of the test set case. Transient state conditions shall meet MIL-PRF-28800 requirements.

2.5 Mean time between failures (MTBF). The equipment MTBF shall be greater than 2000 hours. The lower test shall be 2000 hours and the upper test 4000 hours.

2.6 Pre-conditioning requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn-in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

2.7 Calibration and maintenance adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be for a period of one year or greater based on an operating time of 2000 hours.

2.8 Performance requirements. The instrument shall meet all requirements specified herein, after the warm-up period specified in MIL-PRF-28800, under all combinations of input power conditions, output signal conditions and operating service conditions as specified in MIL-PRF-28800.

2.9 Marker Beacon (MB) Generator.

2.9.1 Marker beacon generator selection. The MB generator shall be selected by means of an ON-OFF control or function selector located on the front panel and labeled as such.

2.9.2 RF frequency. The MB generator shall provide continuous sine waves at the fixed frequency of 75.0 MHz ±0.0025%, and a variable frequency range from 74.6 to 75.4 MHz ±0.0025% in 25 KHz steps. This frequency, referred to as RF herein, shall be modulated as specified herein.

2.9.3 RF frequency stability. The variation in frequency of the RF output frequency shall not exceed ±0.005%.

2.9.4 RF spectral purity. At the frequency specified in 2.9.2, the following specification shall be met:

a. Harmonics on the desired RF signal shall be at least 30 dB below the output level of the desired fundamental RF signal.

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b. Spurious signals, including subharmonics accompanying the desired RF signal, shall be at least 75 dB below the carrier output level at +25.0 KHz in a 300 Hz resolution bandwidth, as measured at the RF IN/OUT connector and into a 50 ohm load.

c. Broadband noise on the desired RF signal shall be at least 115 dB below the carrier output level measured in a 1 Hz noise bandwidth at <40 KHz from the carrier frequency, as measured at the RF IN/OUT connector and into a 50 ohm load.

2.10 RF Output.

2.10.1 RF output attenuator. The RF level shall be adjustable, in not greater than 1 dB increments, from -6 to -120 dBm when measured into a 50 ohm load at the RF IN/OUT connector. RF leakage from cases and cables throughout the specified frequency ranges shall not exceed a level of -115 dBm, measured at the end of the RF IN/OUT cable. The accuracy of the RF output shall be ±1.5 dB for RF output level from -6 to -60 dBm; ±2.0 dB for RF output level from -60 to -120 dBm.

2.10.1.1 RF IN/OUT connector. The RF IN/OUT connector shall be a stainless steel type N female chassis mounting type and be labeled “RF IN/OUT.”

2.10.1.2 RF IN/OUT cable. The RF IN/OUT cable shall be 6 ft ±2 in long RG-213/U or equal cable, with stainless steel type N female connector at one end and a stainless steel type N male connector at the other end.

2.10.1.3 RF output level. The level of the unmodulated RF signal shall be not less than -20 dBm (22.4 mV into 50 ohms) when measured into a 50 ohm load.

2.10.2 Marker tones. The MB carrier shall be amplitude modulated by 400 Hz sine wave (outer marker beacon), 1,300 Hz sine wave (middle marker beacon), and 3,000 Hz sine wave (inner marker beacon) audio frequency (AF) tones.

2.10.2.1 Audio frequency accuracy. The audio signals specified in 2.9.2 shall be accurate to within ±0.005%.

2.10.2.2 Audio frequency spectral purity. For each of the audio signals specified in 2.9.2, the total distortion due to harmonics, noise, and spurious signals shall be less than 1.0%.

2.10.2.3 Modulation depth. Each marker tone shall be capable of amplitude modulating the carrier with a modulation level range from 20 to 95.0%.

2.10.2.3.1 Envelope distortion. The distortion of the modulation envelope on the RF carrier shall be less than 4.0%.

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2.10.2.4 Tone selection. Each tone shall be selected by means of control(s) with each control or controls position properly labeled to indicate the marker beacon frequency or position (i.e., outer, middle, inner).

2.11 Glide Slope Generator.

2.11.1 Glide Slope Generator selection. The glide slope (GS) generator shall be selected by means of an ON-OFF control or function selector located on the front panel and labeled as such.

2.11.2 RF frequency range. The GS generator shall provide continuous sine waves over the frequency range from 329..15 to 335.00 MHz in 150 KHz steps. These frequencies, referred to rf herein, shall be modulated as specified herein.

2.11.2.1 RF frequency selection. The output frequency shall be selected by means of a control located on the front panel. The positions of the control shall be labeled with the appropriate frequency pairs.

2.11.2.2 RF frequency stability. The variation in frequency of the rf output frequency shall not exceed ±0.0025%.

2.11.2.3 RF spectral purity. At the frequencies range specified in 2.11.2, the following specifications shall be met:

a. Harmonics. The level of any harmonic on the desired rf signal shall be at least 25 dB below the output level of the desired fundamental rf signal.

b. Spurious signals: The total of all spurious signals, including subharmonics, accompanying the desired rf signal shall be at least 75 dB below the output level of the desired rf signal at +15 KHz and at least 80 dB below the carrier output level at +25.0 KHz in a 300 Hz resolution bandwidth, as measured at the RF IN/OUT connector and into a 50 ohm load.

c. Broadband noise: The level of any broadband noise on the desired rf signal shall be at least 115 dB below the carrier output level measured in a 1 Hz noise bandwidth at less than 120 KHz from the carrier frequency, as measured at the RF IN/OUT connector and into a 50 ohm load.

2.11.3 Glide slope tones. The GS carriers shall be amplitude modulated by 90 Hz and 150 Hz sine wave af tones.

2.11.3.1 Audio frequency accuracy. The audio signals specified in 2.11.3 shall be accurate to within ±0.005%.

2.11.3.2 Audio frequency spectral purity. For each of the audio signals specified in 2.11.2, the total distortion due to harmonics, noise, and spurious signals shall be less than ±0.25% at preset 40% amplitude modulation (AM) and <±1.0% in the variable range from 10 to 80% AM.

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2.11.3.3 Modulation depth. The 90 Hz and 150 Hz signals shall: be used to amplitude modulate the GS frequencies specified in 2.11.2; be supplied both separately and together; be of sufficient magnitude to modulate the carrier to a preset depth of 40% and a variable range from 10 to 80% AM; vary simultaneously and proportionately.

2.11.3.4 90/150 Hz phase error. The phase error of the 90 and 150 Hz signals shall not exceed 1.0°. Relative phase is defined as the difference in 30 Hz degrees of the points at which the 90 Hz and the 150 Hz wave form crosses the zero axis in the same direction at the same time.

2.11.3.5 DDM control. The output ratio of the 90 Hz signals shall have presets at 0.000, 0.175 and 0.400 ±0.001 DDM, at both 90 Hz predominant and 150 Hz predominant for all settings except 0.000 DDM. For purposes of determining DDM, the base shall be 0.40 for the GS function.

2.11.3.5.1 Variable DDM control. A control shall be provided to continuously vary the DDM from 0.800 ±0.001 DDM, 90 Hz predominant to 0.800 ±0.001 DDM, 150 Hz predominant. This control shall have an off position in which the variable DDM circuitry is not operative.

2.11.3.6 90 Hz and 150 Hz delete. Two front panel controls shall be provided to delete the 90 and 150 Hz tones independently or simultaneously.

2.12 LOCALIZER GENERATOR.

2.12.1 Localizer (LOC) generator selection. The LOC generator shall be selected by means of an ON-OFF control or function selector located on the front panel and labeled as such.

2.12.2 RF frequency range. The LOC generator shall provide continuous sine waves over the frequency range from 108.00 to 117.95 MHz ±0.002% in 50 KHz steps. These frequencies, referred to as rf herein, shall be modulated as specified herein.

2.12.2.1 RF frequency selection. The output frequency shall be selected as specified in 2.12.2.

3.12.2.2 RF frequency stability. The variation in frequency of the rf output frequency shall not exceed ±0.002%.

2.12.2.3 RF spectral purity. At the frequencies specified in 2.12.2, the following specifications shall be met:

a. The level of any harmonic on the desired rf signal shall be at least 30dB below the output level of the desired fundamental rf signal.

b. The total of all spurious signals, including subharmonics, accompanying the desired rf signal shall be at least 75 dB below the output level of the desired rf signal at +15 KHz and at least 80dB below the carrier output level at +25.0 KHz in a 300Hz resolution bandwidth, as measured at the RF IN/OUT connector and into a 50 ohm load.

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c. The level of any broadband noise on the desired rf signal shall be at least 122 dB below the carrier output level measured in a 1 Hz noise bandwidth at 40 KHz from the carrier frequency, as measured at the RF IN/OUT connector and into a 50 ohm load.

2.12.3 Localizer tones. The LOC carriers shall be amplitude modulated by 90 Hz, 150 Hz, and 1020 Hz sine wave af tones.

2.12.3.1 Audio frequency accuracy. The tones in 2.12.3 shall be accurate to within ±0.005%.

2.12.3.2 Audio frequency spectral purity. For each of the audio signals of 2.12.3, the total distortion due to harmonics, noise, and spurious signal shall be <0.25% for the 90 Hz and 150 Hz signals at preset 20% AM, < 1.0% in the variable range from 5 to 40% AM, and <0.5% for the 1020 Hz signal at preset 30% AM.

2.12.3.3 Modulation depth. The 90 Hz and 150 Hz signals: shall be used to amplitude modulate the LOC frequencies specified in 2.12.3; shall be supplied both separately and together; shall be of sufficient amplitude to modulate the carrier to a preset depth of 10% and a variable range from 5 to 40% AM; shall vary simultaneously and proportionately.

2.12.3.4 90 Hz and 150 Hz phase error. The phase error of the 90 Hz and 150 Hz signals shall be as specified in 2.11.3.4.

2.12.3.5 DDM control. The output ratio of the 90 Hz and 150 Hz signals shall have presets at 0.000, 0.046, 0.093, 0.155, and 0.200 ±0.001 DDM, for both 90 Hz dominant and 150 Hz predominant for all signals except 0.000 DDM. For purposes of determining DDM, the base shall be 0.20 when the function selector is in the LOC function.

2.12.3.5.1 Variable DDM control. A control shall be provided to continuously vary the DDM from 0.400 ±0.001 DDM 90 Hz predominant to 0.400 ±0.001 DDM 150 Hz predominant. This control shall have an off position in which the variable DDM circuitry is not operative.

2.12.3.6 90 Hz and 150 Hz delete. Front panel controls shall be provided to delete the 90 Hz and 150 Hz tones independently and simultaneously.

2.12.3.7 ID tone. The 1020 Hz signal shall amplitude modulate localizer frequencies, either separately or with the LOC modulation present and shall be selectable by a front panel ON-OFF control or function selector.

2.12.3.7.1 Modulation. The depth of modulation of the localizer frequencies by the 1020 Hz signal shall be fixed at a modulation level of 30.0% ±2.5% with <1.0% RF tone distortion.

2.13 VHF Omnirange generator.

2.13.1 VOR generator selection. The VOR signal generator shall be selected by means of an ON-OFF control located on the front panel.

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2.13.2 RF frequency range. The VOR signal generator shall provide continuous sine waves over the frequency range from 108.00 to 117.95 MHz ±0.002% in 50 KHz steps. These frequencies, referred to as rf herein, shall be modulated as specified herein.

2.13.2.1 RF frequency selection. The output frequency shall be selected by means of a front panel control.

2.13.2.2 RF frequency stability. The variation in frequency of the rf output frequency shall not exceed ±0.0025%.

2.13.2.3 RF spectral purity. At the frequencies specified in 2.9.2, the following specifications shall be met:

a. The level of any harmonic on the desired rf signal shall be at least 30 dB below the output level of the desired fundamental rf signal.

b. The total of all spurious signals, including subharmonics, accompanying the desired rf signal shall be at least 75 dB below the output level of the desired rf signal at +15KHz and at least 80 dB below the carrier output level at +25.0 KHz in a 300 Hz resolution bandwidth, as measured at the RF IN/OUT connector and into a 50 ohm load.

c. The level of broadband noise on the desired rf signal shall be at least 122 dB below the carrier output level measured in a 1 Hz noise bandwidth at <40 KHz from the carrier frequency, as measured at the RF IN/OUT connector and into a 50 ohm load.

2.13.3 VOR tones. The VOR carriers shall be amplitude modulated by 30 Hz (variable phase), 9960 Hz (frequency modulated by a 30 Hz reference phase signal), and 1020 Hz sine wave af tones.

2.13.3.1 Audio frequency accuracy. The 30 Hz variable and reference phases, the 9960 Hz, and the 1020 Hz signals shall be accurate to within ±0.005%.

2.13.3.2 Audio frequency spectral purity. The total distortion due to harmonics, noise, and spurious signals shall be <0.25% for the 30 Hz reference, 1% for 30 Hz variable, 0.5% for the 1020 Hz identification signal, and 1.5% for 9960 Hz, and 2.0% for auxiliary audio frequencies.

2.13.3.2.1 Incidental AM due to FM. With the 9960 Hz signal frequency modulated with a deviation ratio of 16 ±1 by a 30 Hz signal, the amplitude modulation of the 9960 Hz signal shall not exceed 1.0%.

2.13.3.3 9960 Hz FM subcarrier. The 9960 Hz subcarrier shall be frequency modulated by a 30 Hz fixed reference phase signal, with a peak deviation 480 ±5 Hz. The 9960 Hz FM signal shall amplitude modulate the rf carriers.

2.13.3.4 30 Hz variable phase signal. A second 30 Hz signal shall be provided which is variable in phase with respect to the 30 Hz fixed phase reference signal from 0° to 359.99° lagging in 0.1° steps.

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2.13.3.5 30 Hz frequency equality. The 30 Hz reference signal and 30 Hz variable signal shall be coherent.

2.13.3.6 Bearing signal. The 9960 Hz subcarrier (9960 FM signal), which is frequency modulated by the 30 Hz reference signal, and the 30 Hz variable phase signal shall amplitude modulate the RF carriers both separately and simultaneously.

2.13.3.6.1 Modulation depth. The 9960 Hz FM signal and the 30 Hz variable signal shall each be of sufficient magnitude to amplitude modulate the RF carriers to the depth of preset 30% and variable from 10 to 35%.

2.13.3.6.2 Bearing selector. A control shall be provided to vary the bearing from 000.0 to 359.99° in 10° and 0.1° steps.

2.13.3.6.2.1 Bearing accuracy. The bearing of the VOR signal on the RF carrier shall be within ±0.14° of the audio bearing as set by a VOR standard (Collins Radio Group model 478A.3 or equal).

2.13.3.7 30 Hz variable and 30 Hz reference delete. Front panel control shall be provided to delete the 30 Hz variable and 30 Hz reference signals independently and simultaneously.

2.13.3.8 ID tone. The 1020 Hz shall amplitude modulate VOR frequencies both separately an with the VOR modulation present. A ON-OFF front panel control shall be provided for this signal.

2.13.3.8.1 Modulation. The depth of modulation of the VOR frequencies by the 1020 Hz signal shall be fixed at a modulation level of 30.0% ±2.5% with less than 1% RF distortion.

2.14 Interface. The VOR/ILS Signal Generator shall provide a IEEE Standard 488 interface function and connector. The signal generator shall have a capability of being remotely programmed all the necessary functions (LOC, MB, GS, VOR) and performing a store and recall function.

2.15 Self test. The VOR/ILS Signal Generator shall have a self test function.

2.16 Dimensions and weight. The size and weight of the signal generator shall be consistent with current commercial capabilities. Approximate dimensions should be 10 inches high x 19 wide x 22 deep. The weight, including all accessories and excluding manuals, shall not exceed 45 lb.

3. REGULATORY REQUIREMENTS

3.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

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4.1 Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Market Acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID.

4.2.1 The product offered must have been sold to the government or commercial market.

4.2.2 Bid samples shall be required for customized products or those without a history of market acceptability. The product shall be bid sample tested within the accuracies, limits and specifications herein

4.2.3 The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING

5.1 Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES

6.1 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using the conversion tables contained in the latest revision of FED-STD-376 and all other requirements of this CID are met. If a product is manufactured to metric dimensions exceeding the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

6.2 Source of Documents.

6.2.1 The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.2.2 ANSI Standards are available from the American National Standards Institute, 1430 Broadway, New York, NY 10018.

6.2.3 Federal and Military specifications are available from the Defense Documents Order Desk, Building 4D, 700 Robins Ave, Philadelphia, PA 19111-5094.

6.3 National Stock Numbers (NSNs). The following is a list of NSNs which correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID:

6625-01-075-1814	622-4127-001	Rockwell
6625-01-381-1329	2031	Marconi

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6.4 Part Identification Number (PIN). National Stock Numbers of products meeting the requirements of this CID will be assigned part number AA59127 in the Federal Cataloging System. This action is for government purposes and does not constitute a requirement for the contractor.

MILITARY INTERESTS:

Custodian:
Air Force - 99

Review Activities:
Air Force - 01
OSD - SO

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

Preparing Activity:
Air Force - 82

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